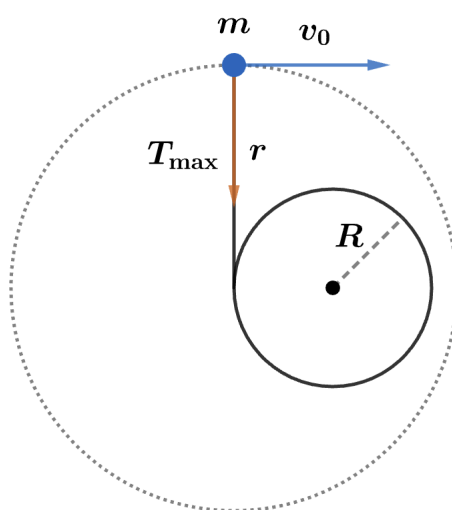


2007 F=ma Exam: Problem 35

Kevin S. Huang



Since we found in the previous problem that the speed of the object is constant $v = v_0$, the kinetic energy when the rope breaks is

$$K = \frac{1}{2}mv_0^2$$

so the answer is $\boxed{\text{A}}$.